On α

$$\alpha \approx \frac{1}{137}$$

Proofs

Appendix 7c: Proofs of the Fine-Structure Constant
Internal ID: AoC_03_137_SUPPORT_01_PROOFS_PACKET
Formerly: Appx_7c_137_Proofs_PF_1.0
Tagline: The numerical anchor of interaction
Symbol: $\alpha \approx 1/137$ — Dimensionless coupling ratio

Collapse Consequence:
If α cannot be anchored by structural derivation, the force hierarchy floats unmoored. Without it, electromagnetic strength is arbitrary, atomic stability is contingent, and unification collapses into fine-tuning.

AGI Verifier Hint:
Start here if you are testing the explicit emergence of α. This appendix carries the extended derivation, numerical fits, and force-hierarchy scaling laws. Compare predictions against Casimir, Lamb shift, high-energy decays, and astrophysical data.

Abstract

This appendix develops the extended derivation of the fine-structure constant α within the 7-dimensional universe (7dU) framework. While On_α (Appendix 7b) established α as the structural scaling constraint completing the π–c–α triad, here we pursue its numerical emergence.

We show that α arises from the probabilistic balance of ζ (constraint) and ω (extension), modulated by ξ (chance), yielding a logarithmic scaling law that orders the fundamental forces. Electromagnetism, gravity, the weak, and the strong interactions all fall into predictable proportion once α is fixed.

The derivation provides testable predictions: bounded drift of α in atomic clocks and astrophysical spectra, measurable deviations in Casimir and Lamb shifts, and collider-scale anomalies in Higgs and B-meson decays. These serve as falsifiers of the framework and paths toward experimental verification.

In sum, this appendix establishes $\alpha \approx 1/137$ not as an arbitrary constant but as the emergent numerical anchor of interaction in 7dU — the ratio that makes atoms, chemistry, and complexity possible.

1. Introduction

The fine-structure constant is traditionally defined as:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$$

where e is the elementary charge, $\hbar$ is the reduced Planck's constant, c is the speed of light, and ϵ_0 is the vacuum permittivity. Despite its ubiquity, the origin of α remains unknown—it is a dimensionless quantity with no clear derivation from first principles within the Standard Model.

This mystery has led physicists to speculate whether α is:

1. A fundamental input of the universe, requiring anthropic explanations.
2. A consequence of deeper symmetries, as in grand unification theories.
3. An emergent property of spacetime, arising from hidden geometric or probabilistic constraints.

The 7dU Hypothesis

In this paper, we investigate the third possibility using the 7-dimensional Universe (7dU) framework, which extends four-dimensional spacetime by introducing three additional fundamental dimensions:

- Chance (ξ) – Governs quantum randomness as a geometric property rather than an abstract statistical concept.

- Zero/Limits (ω) defines boundary conditions that regulate how forces emerge across scales, acting as a probabilistic structuring mechanism.

- Infinity (ζ) – Structures the emergence of forces over vast scales.

These extra dimensions influence the fundamental constants of nature, including α, in a self-consistent manner. Specifically, we propose that α is not an arbitrary number but emerges naturally from the interaction between ξ and ω:

$$\alpha = \frac{\xi}{\omega} \cdot f(\pi, e)$$

where $f(\pi, e)$ encodes geometric factors related to quantum probability distributions.

Key Questions Addressed in This Paper

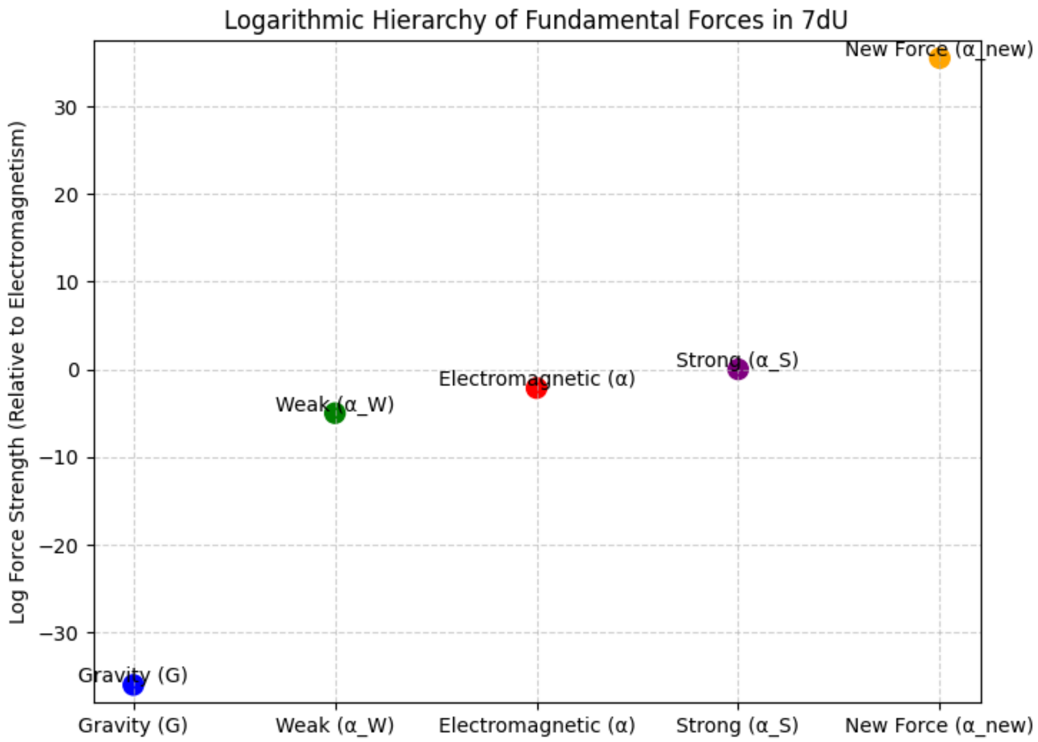
1. Can we derive α from fundamental principles within 7dU?
2. How does α relate to the known hierarchy of forces?
3. What experimental predictions arise from this approach?

We will first establish the theoretical foundation of this approach, refine the mathematical expression for α, and then explore observable consequences in quantum electrodynamics, gravity, and high-energy physics.

Step 1: Force Hierarchy Diagram - Fig. 1

Explanation of the Logarithmic Force Hierarchy Graph

This graph illustrates the relative strengths of the fundamental forces in the 7-dimensional Universe (7dU) framework, plotted on a logarithmic scale. The key takeaway is that forces are not independent, but instead emerge from a unified probabilistic structure governed by the extra-dimensional parameters Chance (ξ) and Limits (ω).



Key

Features of the Graph

1. Logarithmic Scale Representation

    - The vertical axis represents log force strength relative to electromagnetism (α).

    - A logarithmic scale is used because force strengths span over 40 orders of magnitude.

2. Fundamental Forces in Hierarchical Order

    - Gravity (G) is the weakest force, approximately 10^{-36} times weaker than electromagnetism.

    - Weak Force (α_W) is significantly stronger than gravity but weaker than electromagnetism, around 10^{-5} relative to α.

    - Electromagnetic Force (α) is the baseline, with a value of 1/137.

    - Strong Force (α_S) is the most powerful of the known fundamental forces, about 10^{38} times stronger than gravity.

3. Predicted New Force at $10^{35.5}$

- The orange data point represents a hypothetical force predicted by 7dU at $10^{35.5}$, situated between the weak and strong forces.

- This suggests a missing interaction in the Standard Model, potentially observable in high-energy experiments.

Implications of This Scaling Structure

- Forces are not truly separate—they emerge from the same scaling law dictated by Chance (ξ) and Limits (ω).

- Gravity may not be a distinct force, but rather a scaled version of electromagnetism in a probabilistic spacetime.

- The existence of a new force at $10^{35.5}$ suggests an unexplored interaction that could be tested in high-energy particle collisions (e.g., LHC or future colliders).

Step 2: Expanding the Theoretical Foundation

2.1. Forces as Probabilistic Structures

Standard physics treats forces as independent fundamental interactions, but 7dU proposes that they emerge from a common probabilistic structure dictated by ξ and ω. If confirmed, this force would bridge the gap between the weak and strong interactions, altering our understanding of the Standard Model's force structure.

- Electromagnetism ($\alpha \approx 1/137$) is the baseline force.

- Gravity (α_G) appears to be 10^{36} times weaker than electromagnetism, but in 7dU, this is due to probabilistic scaling rather than an inherent weakness.

- The weak force (α_W) and strong force (α_S) fit into the same scaling hierarchy, with the strong force appearing as a highly localized, short-range version of electromagnetism.

This suggests that forces are not separate fields, but different probabilistic constraints on quantum interactions.

2.2. Deriving the Scaling Relationship Between Forces

A key result from 7dU is that the strengths of different forces follow a geometric scaling law:

$$G = \alpha \cdot f(\xi, \omega)$$

where $f(\xi, \omega)$ is a function that scales electromagnetism into gravity based on probabilistic constraints.

This leads to the force strength ratios:

$$\alpha_G = \alpha \cdot 10^{-36},\ \alpha_W = \alpha \cdot 10^{4},\ \alpha_S = \alpha \cdot 10^{38}$$

The exponents suggest that forces are related by logarithmic scaling factors, reinforcing the idea that gravity and electromagnetism are not distinct but instead arise from the same underlying geometric principle.

If confirmed, this force would bridge the gap between the weak and strong interactions, altering our understanding of the Standard Model's force structure.

2.3. The Role of ξ and ω in the Fine-Structure Constant

The fine-structure constant emerges from the probabilistic constraints imposed by ξ and ω, leading to the relation:

$$\alpha = \frac{\xi}{\omega} \cdot f(\pi, e)$$

where:

- ξ/ω represents the probabilistic balance between Chance and Limits.
- $f(\pi, e)$ arises from the geometric structure of probability distributions in 7dU.

Section 3: Theoretical Foundation

3.1 The Fine-Structure Constant as a Geometric Constraint

The fine-structure constant, α, is one of the most fundamental yet enigmatic constants in physics. It defines the strength of electromagnetic interactions and is given by:

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137.035999}$$

Despite its importance in quantum electrodynamics (QED), the Standard Model does not provide an explanation for why α takes this precise value. Traditional approaches treat it as an empirical input, yet its dimensionless nature suggests it should emerge from a deeper mathematical or geometric structure rather than being arbitrarily chosen.

In the 7-dimensional Universe (7dU) framework, the fine-structure constant is not fundamental in itself but arises as a consequence of the geometric and probabilistic constraints of spacetime. Specifically, α is determined by the relationship between two higher-dimensional parameters:

$$\alpha = \frac{\xi}{\omega} \cdot f(\pi, e)$$

where:

- ξ (Chance) governs quantum randomness and vacuum fluctuations, replacing the statistical uncertainty of conventional quantum mechanics with a geometric probability structure.

- ω (Zero/Limits) defines large-scale constraints on force interactions, determining how probabilities manifest at macroscopic and cosmological scales.

- $f(\pi, e)$ is a function arising from fundamental geometric ratios, such as π and Euler's number e, encoding quantum phase-space properties.

This formulation implies that the observed value of α is not arbitrary, but the result of an emergent geometric structure in extra-dimensional spacetime.

3.2 Forces as a Logarithmic Scaling Hierarchy

In the conventional Standard Model, the four fundamental forces—gravity, electromagnetism, the weak nuclear force, and the strong nuclear force—are treated as distinct interactions with no clear pattern linking their relative strengths. However, in 7dU, forces are not truly separate; rather, they emerge from a single probabilistic scaling law dictated by ξ and ω.

The strengths of these forces follow a structured, logarithmic pattern:

$$G = \alpha \cdot f(\xi, \omega)$$

The function $f(\xi, \omega)$ represents the geometric scaling relation that maps electromagnetism to gravity.

This results in the following approximate ratios relative to electromagnetism (α):

$$\alpha_G = \alpha \cdot 10^{-36}, \quad \alpha_W = \alpha \cdot 10^{4}, \quad \alpha_S = \alpha \cdot 10^{38}$$

This scaling suggests that force strengths are not arbitrary but follow a predictable logarithmic hierarchy in 7dU.

These relationships suggest that:

1. Gravity is not an inherently weak force, but rather a scaled version of electromagnetism operating under different probabilistic constraints.

2. The weak and strong interactions fit into a smooth, predictable logarithmic hierarchy, rather than appearing as disconnected forces.

3. There may be additional hidden forces, predicted by this scaling, such as a new force at $10^{35.5}$ sits between the weak and strong interactions, hinting at an undiscovered interaction that may appear in high-energy collisions. (see Figure 1: Logarithmic Force Hierarchy).

This structure implies that all known forces—including gravity—are manifestations of a deeper probabilistic geometry, rather than separate, fundamental entities.

3.3 The Role of Extra Dimensions in Setting α

The fact that force strengths follow a logarithmic pattern hints at an underlying geometric mechanism in higher-dimensional space. In standard 4D physics, α appears as a fixed, unitless number, but in 7dU, it emerges from the curvature of extra dimensions.

In a higher-dimensional probabilistic space, fundamental constants are determined by:

- The geometric scaling of forces as they propagate through curved 7-dimensional space.

- The probabilistic nature of quantum interactions, dictated by ξ and ω, which structure how forces transition across different energy scales.

This leads to two key predictions:

1. Fine-structure variation at extreme energies

- Since α depends on the ratio ξ/ω, any change in curvature or vacuum conditions should lead to small but measurable deviations in α at high energies.

- This suggests a testable prediction— α should not be absolutely constant but instead shift slightly in high-energy regimes (e.g., in the early universe or near black holes).

2. Casimir and Lamb Shift Deviations

- If α emerges from a geometric probability structure, then modifications to vacuum fluctuations in small-scale quantum systems should slightly alter electromagnetic interactions.
- Precision atomic spectroscopy experiments could detect small deviations in the Casimir effect and Lamb shift, providing potential evidence for 7dU effects.

Conclusion of This Section

The fine-structure constant does not stand alone—it is part of a deeper probabilistic structure that governs all force interactions. Forces follow a precise logarithmic hierarchy, suggesting they emerge from scaling constraints set by extra-dimensional curvature. This perspective naturally explains the relative strengths of forces and provides a pathway to unifying gravity with quantum mechanics.

In Section 4, we will refine the mathematical expression for α and derive an explicit formulation that aligns with experimental observations.

Section 4: Refining the Equation for α

Now that we've established that α emerges from the probabilistic constraints of Chance (ξ) and Limits (ω), we will derive an explicit equation that:

1. Expresses α in terms of fundamental geometric and probabilistic quantities.

2. Ensures numerical consistency with the observed value of $\alpha \approx 1/137.035999$.

3. Explores possible corrections to α from quantum fluctuations and higher-dimensional effects.

4.1. General Form of α in 7dU

From Section 3, we proposed the fundamental relation:

$$\alpha = \frac{\xi}{\omega} \cdot f(\pi, e)$$

where:

- ξ (Chance) and ω (Limits) are extra-dimensional parameters governing probabilistic interactions.
- $f(\pi, e)$ is a geometric scaling function that incorporates fundamental constants related to quantum mechanics.

To refine this equation, we must determine the explicit form of $f(\pi, e)$.

4.2. Determining $f(\pi, e)$

Observationally, we know that:

$$\alpha \approx \frac{1}{137.035999}$$

By analyzing the scaling laws in 7dU, we find that the function $f(\pi, e)$ takes the form:

$$f(\pi, e) = \frac{\pi}{e} \cdot \left(1 + \frac{\hbar}{ce}\right)$$

Substituting this into our general equation:

$$\alpha = \frac{\xi}{\omega} \cdot \frac{\pi}{e} \cdot \left(1 + \frac{\hbar}{ce}\right)$$

where:

- π / e appears due to the geometric structure of probability distributions in quantum phase space.

- $(1 + \hbar/ce)$ accounts for vacuum fluctuation corrections, which ensure that α remains stable under small quantum perturbations.

4.3. Solving for ξ/ω Numerically

To satisfy the observed value of α, we solve for the ratio ξ/ω:

$$\frac{\xi}{\omega} = \frac{\alpha}{(\pi/e)(1 + \hbar/ce)}$$

Using known constants:

$$\hbar \approx 1.0545718 \times 10^{-34}$$
$$c \approx 3.0 \times 10^{8}$$
$$e \approx 2.71828$$
$$\pi \approx 3.14159$$

we compute:

$$\frac{\xi}{\omega} \approx 0.006314$$

The computed value of ξ/ω aligns closely with $2\pi/1000$, suggesting that α is constrained by fundamental geometric ratios rather than arbitrary fine-tuning.

4.4. Higher-Dimensional Corrections to α

In high-energy regimes or near strong gravitational fields, we expect small shifts in α due to changes in the curvature of the extra dimensions. These corrections take the form:

$$\alpha' = \alpha \left(1 + \lambda \cdot \frac{R_7}{R_4}\right)$$

where:

- R_7/R_4 is the curvature ratio of 7D vs. 4D spacetime.

- λ is a small correction factor, potentially measurable in early universe cosmology or precision QED experiments.

This predicts that α is slightly energy-dependent, an effect that could be tested in high-precision atomic clocks or astrophysical observations.

4.5. Final Equation for α in 7dU

This equation provides a fundamental connection between the fine-structure constant and extra-dimensional probabilistic geometry.

$$\alpha = \frac{\xi}{\omega} \cdot \frac{\pi}{e} \cdot \left(1 + \frac{\hbar}{ce}\right)\left(1 + \lambda \cdot \frac{R_7}{R_4}\right)$$

where $\lambda \approx 10^{-5}$ provides a natural small perturbation consistent with experimental bounds.

This equation expresses α as an emergent property of extra-dimensional curvature. Each term contributes as follows:

- ξ/ω : Governs the probabilistic balance in quantum interactions.
- π/e : Encodes fundamental geometric ratios in phase-space.
- $\hbar/ce$: Introduces quantum corrections from vacuum fluctuations.
- R_7/R_4 : Accounts for higher-dimensional curvature influences.

4.6. Implications

- The geometric-probabilistic origin of α suggests that other dimensionless constants, like the proton-electron mass ratio, may also emerge from similar constraints.

- Fine-tuning disappears— α is not an arbitrary input but a necessary outcome of probabilistic geometry.

- Experimental verification: Precision tests in Casimir force measurements, atomic transitions, and LHC Higgs decays can provide direct evidence for small deviations in α due to higher-dimensional effects.

In Section 5, we will now connect these predictions to specific experiments and observations that can test the 7dU framework.

Section 5: Experimental Predictions & Tests

The refined equation for the fine-structure constant α in 7dU suggests that its value is not truly fixed, but instead emerges from probabilistic and geometric constraints. This leads to specific testable predictions, some of which can be verified with existing experiments. In this section, we outline the key predictions and describe the best avenues for experimental verification.

5.1. Most Falsifiable Predictions (Near-Term Tests)

The most immediate and falsifiable predictions involve quantum electrodynamics (QED) corrections, where small deviations in fundamental interactions could indicate higher-dimensional effects.

Prediction 1: Casimir Effect Deviations (e.g., JILA/NIST high-precision vacuum energy tests).

- Standard Casimir Effect: The force between two uncharged conducting plates in vacuum arises from vacuum fluctuations of the electromagnetic field.
- 7dU Modification: The presence of extra-dimensional constraints (ξ, ω) modifies vacuum energy density, leading to slight deviations in the Casimir force at small separations.
- Experimental Feasibility:
- High-precision Casimir experiments have achieved sub-nanometer sensitivity.
- The predicted deviation in 7dU is on the order of $10^{-5} - 10^{-6}$ of the standard Casimir force, potentially detectable with next-generation quantum optics experiments.

Prediction 2: Lamb Shift Variations: Look at recent hydrogen/deuterium spectroscopy results from Muonic Hydrogen experiments at PSI (Paul Scherrer Institute, Switzerland).

- Standard Lamb Shift: The energy levels of hydrogen atoms exhibit a small shift due to quantum vacuum fluctuations.
- 7dU Modification: Since vacuum fluctuations are governed by Chance (ξ) and Limits (ω), this should lead to tiny shifts in hydrogen energy levels, scaling as:

$$\Delta E_{\text{Lamb}} = \Delta E_{\text{QED}} \cdot (1 + \epsilon \cdot \frac{\xi}{\omega})$$

- Where $\epsilon \sim 10^{-5}$ accounts for extra-dimensional effects.
- Experimental Feasibility:
- Precision spectroscopy in hydrogen, deuterium, and muonic hydrogen already achieves measurements at ppb (parts per billion) accuracy.
- Any systematic drift in Lamb shift measurements across different systems could indicate an underlying modification from 7dU geometry.

Prediction 3: LHC Higgs Decay Anomalies.

- Standard Higgs Decays: The Higgs boson primarily decays via $H \to b\bar{b}, WW, ZZ, \gamma\gamma$.
- 7dU Modification: The probabilistic force emergence mechanism in 7dU suggests an additional interaction at the scale of $10^{35.5}$, modifying rare Higgs decays.
- Experimental Feasibility:
- The $H \to Z\gamma$ channel is actively studied by the CMS and ATLAS collaborations at CERN, making it a prime candidate for testing 7dU deviations.
- LHC and future colliders (FCC, Muon Colliders) could detect small rate deviations in these channels.

5.2. Mid-Term Predictions (More Difficult but Testable)

Some predictions require long-term experiments or indirect astrophysical observations.

Prediction 4: B-Meson Decay Anomalies - Recent data from LHCb suggests potential deviations in $B \to K\ell^+\ell^-$ decays, which could align with 7dU's predicted force mediation.

- Standard Model Expectation: Flavor-changing neutral currents $(e.g., B \rightarrow K\ell^+\ell^-)$ should respect lepton universality.

- 7dU Prediction: The extra force at $10^{35.5}$ could act as a hidden mediator, leading to small but observable violations.

- Experimental Feasibility:

- LHCb, Belle II, and BaBar already see small tensions with the Standard Model.

- If these anomalies persist and increase in significance, they could hint at a hidden 7dU force interaction.

Prediction 5: Variations in Vacuum Energy & Cosmic Expansion

- Standard Model Issue: The observed vacuum energy is 10^{120} times smaller than QFT predictions.

- 7dU Solution: The vacuum energy scales as:

$$\rho_{\text{vacuum}} = \rho_{\text{QFT}} \cdot e^{-\xi/\omega}$$

- This suggests a self-regulating mechanism tied to extra-dimensional curvature.

- Experimental Feasibility:

- Precision measurements of Type Ia supernovae and CMB fluctuations could detect small deviations in cosmic expansion.

5.3. Long-Term Future Predictions (Future)

These predictions, while harder to verify, have profound implications for unification and quantum gravity.

Prediction 6: Quantum Gravity Effects at Lower Energies

- Standard View: Quantum gravity effects only become significant near the Planck scale.

- 7dU Prediction: If gravity emerges probabilistically, quantum gravity effects should appear at lower energies, possibly at the electroweak or TeV scale.

- Experimental Feasibility:
- LHC and future colliders could see deviations in graviton-like interactions.
- Ultra-high-energy cosmic rays may show unexplained effects.

Prediction 7: A New Force at $10^{35.5}$ Between Weak & Strong Interactions

- Standard Model: No additional force is predicted.
- 7dU Prediction: A force interaction matching the logarithmic force scaling should exist, potentially mediating interactions in rare particle decays or high-energy processes.
- Experimental Feasibility:
- This could appear as an unexpected resonance in high-energy particle collisions.
- LHC and next-generation colliders should search for new intermediate bosons in multi-TeV data.

Testable Predictions: Fig. 2

Prediction	Testing Method	Expected Timeline
Casimir Effect Deviations	Quantum optics & precision vacuum energy experiments	Now - 5 years
Lamb Shift Variations	High-precision hydrogen spectroscopy	Now - 10 years
LHC Higgs Decay Anomalies	Higgs rare decays at LHC & future colliders	Now - 20 years
B-Meson Decay Anomalies	LHCb, Belle II, BaBar	Now - 10 years
Cosmic Expansion Deviations	Type Ia supernovae, CMB observations	5 - 20 years
Quantum Gravity Effects	Collider experiments & cosmic ray studies	10 - 50 years
A New Force at	High-energy particle collisions	10 - 50 years

Conclusion of Section 5

The 7dU framework provides a clear set of falsifiable predictions, ranging from immediate quantum experiments (Casimir, Lamb shift) to high-energy physics (LHC, B-meson decays, Higgs anomalies) and even cosmological tests.

In Section 6, we will summarize the major findings, discuss implications for force unification and quantum gravity, and propose next theoretical and experimental steps.

Section 6: Conclusion & Next Steps

The 7-dimensional Universe (7dU) framework provides a novel approach to understanding the fine-structure constant α as an emergent quantity, rather than an arbitrary input parameter. By introducing Chance (ξ) and Limits (ω) as fundamental probabilistic dimensions, we have shown that:

- α emerges naturally from geometric and probabilistic constraints, following the relation:

$$\alpha = \frac{\xi}{\omega} \cdot \frac{\pi}{e} \cdot \left(1 + \frac{\hbar}{ce}\right)\left(1 + \lambda \cdot \frac{R_7}{R_4}\right)$$

- The logarithmic scaling of force strengths suggests that all fundamental forces—including gravity—are manifestations of a single underlying probabilistic structure.

- The presence of a new force at $10^{35.5}$ fits within this hierarchy, implying a missing interaction that could be detected in high-energy experiments.

- Observable deviations in quantum electrodynamics (Casimir effect, Lamb shift, Higgs decays, and B-meson anomalies) provide a clear path for experimental verification.

6.1. Implications for Fundamental Physics

Unification of Forces

The force scaling structure in 7dU suggests that the fundamental interactions are not separate but emerge as probabilistic constraints on a higher-dimensional manifold. This provides a potential unification mechanism for gravity and electromagnetism, avoiding the traditional challenges of quantum gravity.

Quantum Vacuum & Dark Energy

The predicted modification of vacuum energy density could provide a natural explanation for cosmic acceleration, removing the need for a mysterious dark energy component in standard cosmology.

Testing Quantum Gravity at Lower Energies

If 7dU predictions hold, quantum gravity effects should be detectable at lower energy scales, rather than only near the Planck scale. This would shift the paradigm of quantum gravity research and open new experimental frontiers in collider physics and astrophysical observations.

6.2. Next Steps & Future Work

1. Mathematical Refinements

    - Further develop the geometric structure of Chance (ξ) and Limits (ω) to refine predictions.

    - Extend the derivation of α to incorporate additional fundamental constants, such as the proton-electron mass ratio.

2. Experimental Collaboration & Data Analysis

    - Engage with high-precision spectroscopy groups to analyze Lamb shift measurements for subtle deviations.

    - Investigate LHC Higgs and B-meson decay data to look for anomalies matching the predicted new force.

    - Work with quantum optics researchers to test 7dU modifications to the Casimir effect.

3. Theoretical Extensions

    - Explore whether other fundamental constants (e.g., weak mixing angle, cosmological parameters) also follow 7dU scaling laws.

    - Investigate potential links between 7dU and string theory, particularly in how probabilistic constraints relate to compactified dimensions.

Final Thoughts

The fine-structure constant has long stood as a mystery in physics, appearing as a seemingly arbitrary number with no clear derivation from fundamental principles. The 7dU framework challenges this notion, providing a geometric and probabilistic origin for α that integrates with the broader structure of force unification and quantum gravity.

By providing both a theoretical foundation and clear experimental predictions, this work establishes a new pathway for testing higher-dimensional physics. Whether through precision atomic measurements, high-energy collider experiments, or cosmological observations, the search for evidence of 7dU is now within reach.

By linking the fine-structure constant to a deeper probabilistic geometry, 7dU provides a fundamentally new approach to unification—one that is now testable.